Giovani Diaz Alfaro

Human Factors & UX Researcher

Human factors researcher with 6 years of experience studying how people work with complex systems in aerospace, defense, and healthcare. Experienced in designing and running mixed-methods studies (experiments, surveys, interviews, task analysis) and turning the findings into design recommendations that engineering and product teams adopt.

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EDUCATION

MSc Interdisciplinary Studies (conc. Cognitive Systems, Ethics)
University of Central Florida
Fall 2026

GradCert. Cognitive Sciences
University of Central Florida
Spring 2023

BA Philosophy (min. Cognitive Sciences)
University of Central Florida
Spring 2020

SOFTWARE

Scripting and Statistics: Python, R, MATLAB, SQL (basic)

Research Tools: Qualtrics, Miro, Empatica

Data and Reporting: Excel, Epic EHR

Webdev: JavaScript, WordPress, basic HTML5

Publishing: LaTeX

LANGUAGES

Bilingual: English, Spanish

Basic Proficiency: French

PROFESSIONAL EXPERIENCE

Researcher
Cognitive Sciences Laboratory, University of Central Florida
2023 – 2026

Lockheed Martin Corporation: Externalized and Extended Cognition, *Cognitive Offloading for Manned-Unmanned Teaming*

- Built a design framework on top of an existing cognitive task analysis of Lockheed Martin's Future Vertical Lift environment, separating cognitive offloading from scaffolding as two independent levers for managing pilot workload.
- Produced three testable design recommendations that were adopted into Lockheed's engineering design process, and designed the experiments to test them. First-author paper, HFES 2024.
- Scoped the problem in working sessions with Lockheed and partner engineering teams, mapping their goals and constraints onto research questions.
- Findings informed a UI overhaul of an active operations monitoring and information management system.

U.S. Department of Defense, NASA, Minerva Research Initiative: *Understanding Cognition in Support of Future Space Missions*

- Helped develop unobtrusive multi-sensor measures (physiological and environmental) of individual and team cognitive states for long-duration space missions.
- Contributed workflow and environment design recommendations for long-duration missions with human and AI (onboard system) team members.

- Designed and ran experiments (including a VR study on Oculus), usability sessions, Qualtrics surveys, and participant and end-user interviews, and analyzed the results in R and Python (ANOVA, regression, factorial designs).
- Coded qualitative team communication data and calibrated coding schemes across raters using inter-rater reliability checks.
- Managed the lab team and human-subjects logistics, including participant and interview scheduling, IRB protocols, and coordination with partner labs at other universities.
- Wrote research proposals for funding and presented findings to engineers, sponsors, and academic audiences.

Supervisor, Process Improvement Coordinator
Orlando Health Imaging Centers
2020 – 2022
Project(s): Quality Assurance Program, St. Petersburg Expansion, Epic Systems Launch and Training

- Conducted task analysis across clinical and non-clinical workflows to find failure points and inefficiencies, then defined performance metrics and led organization-wide process redesigns to reduce error and improve safety.
- Analyzed operational data against internal and external benchmarks to identify human-error risk points in clinical workflows. Designed and implemented process interventions that reduced errors by 60% and

KEY SKILLS

Research Methods:
Usability testing, cognitive task analysis, experimental and survey design, participant and end-user interviews, mixed methods, human-subjects research and IRB

Analysis:
ANOVA, regression, factorial designs, hypothesis testing, qualitative coding, inter-rater reliability

Collaboration: Cross-functional and multi-institution research, stakeholder communication, public speaking

Other: Team and staff management, computer hardware

CERTIFICATIONS

Python 3 (Codecademy, 2022)

Inclusive and Ethical Leadership Certificate (Leadership Summit & USF, 2024)

RESEARCH INTERESTS

- Human-Machine Teaming
- Practical Cognitive Science
- Team Cognition
- Interdisciplinarity
- Software for Scientific Teams
- Human Enhancement
- Moral Cognition
- Ethics of Artificial Intelligence
- Critical Pedagogy

increased departmental productivity by 40%.
- Supervised staff and led cross-functional teams through training on redesigned clinical workflows.
- Monitored operational data trends to surface workflow improvements, and reported productivity, error, and revenue metrics across departments in Excel to support evidence-based process decisions.
- Ran ongoing process audits and workflow diagramming, presenting findings to build stakeholder buy-in for changes.
- Managed training and built reference resources for the organization-wide Epic EHR launch.

PROFESSIONAL DEVELOPMENT

Applied Machine Learning: CNN Image Classifier (CAP 5610, UCF, 2024)
Co-built and trained a VGG-16 convolutional neural network classifying 130 classes (about 6,100 images). Handled data cleaning, augmentation, and epoch tuning, reaching 78.5% accuracy, and benchmarked against comparable community models.

Team Learning for Better Scientific Software (TL4BSSw)
Newton, O. B., **Diaz Alfaro, G.**, et al. Team learning for better scientific software. Manuscript under revision, IEEE Computing in Science and Engineering.

RELATED EXPERIENCE

Guest Lecturer
Topics in Cognitive Science (Fall '24, '25) and *Research Methods in Cognitive Science* (Spring '25)
UCF graduate courses, ≤15 students

Teaching Assistant
Ethics in Science & Technology (Spring '20), UCF

Reviewer
Human Factors and Ergonomics Society International Annual Meeting (2024 – 2026)

PUBLICATIONS

Conference Proceedings
Diaz Alfaro, G., Fiore, S. M., & Oden, K. (2024). Externalized and extended cognition: Cognitive offloading for human-machine teaming. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting.*
https://doi.org/10.1177/10711813241275506

Presentations, Panels, and Posters
Diaz Alfaro, G. (2024, November). Team Learning for a Better Foundation for Scientific Software. Position Paper presented at the 2024 Team Learning for Better Scientific Software Workshop, Orlando, FL.

Sonnenfeld, N., Nguyen, B., **Diaz Alfaro, G.**, Daniliv, V., Newton, O. B., Jentsch, F., Fiore, S. M. (2024). Eliciting Requirements and Recommendations for Training Macrocognition in Teams: Considerations for Collaborative Problem Solving. *Poster presented at 2024 ASPIRE, HFES International Annual Meeting, Phoenix, AZ.*